Google Cloud

Digital Transformation with Google Cloud

This study guide covers the materials from the first course of the six-course series for the Cloud Digital Leader certification.

Certification exam focus

The Cloud Digital Leader certification exam is designed to assess knowledge of cloud literacy, common business use cases, and how cloud solutions and the capabilities of Google Cloud products support an enterprise.

What is the cloud?

The cloud refers to the on-demand delivery of computing power, database storage, apps, and other IT resources via the internet with pay-as-you-go pricing.

What is digital transformation?

When an organization uses modern digital technologies such as cloud platforms to create or modify business processes, culture, and customer experiences to meet changing business and market dynamics, it's considered digital transformation.



The common drivers of digital transformation include:

- Staying ahead of competitors.
- Avoiding integration tax.
- New security challenges.
- The need for a connected architecture and flexible solutions.

Cloud deployment models and strategies

Models:

- **On-premises:** When an organization hosts and manages its own apps and data on servers located in its own data center or facility.
- **Private cloud:** Cloud computing resources used exclusively by one organization. Resources can be physically located on the organization's site (on-premises) or managed by a third-party provider.
- **Public cloud:** Cloud services offered by a third-party provider (like Google Cloud) over the public internet. These resources are shared by multiple organizations or users.

Strategies:

- **Hybrid cloud:** Infrastructure environment that combines a public cloud and a private cloud (or on-premises infrastructure), allowing data and apps to be shared between them.
- **Multicloud:** The use of cloud services from more than one different public cloud provider.

Benefits of cloud technology



Scalability



Agility



Cost-effectiveness



High availability



Strategic value



Flexibility



Speed



Global reach



Data-driven insights



Security

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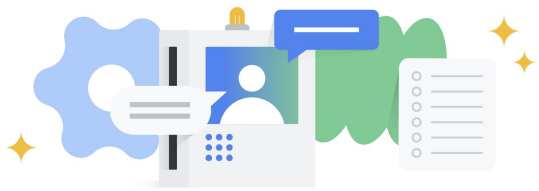
Digital Transformation with Google Cloud

Google Cloud's strengths



- World-leading AI
- Openness and interoperability
- AI Hypercomputer
- AI-optimized data platform
- Security
- Global network

Agentic AI: Refers to AI systems that can independently plan, use tools, and interact with other systems to complete complex, multistep workflows.



Cloud computing service models

Infrastructure as a Service (IaaS)

Similar to leasing a car, IaaS provides on-demand access to core resources like compute and storage without the burden of owning hardware. It offers high scalability and a pay-as-you-go model.

Example: Compute Engine, Cloud Storage

Platform as a Service (PaaS)

Operating like a taxi service, PaaS provides a complete environment for developers to build and run applications while the provider manages patches and infrastructure.

Example: Gemini Enterprise Agent Platform

Software as a Service (SaaS)

Comparable to riding a bus, SaaS delivers complete, ready-to-use applications directly through a web browser. It eliminates the need for local installation or maintenance, providing a cost-effective, subscription-based solution.

Example: Google Workspace

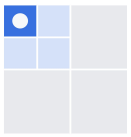
Networking terms and concepts

- **IP address:** Unique numerical label that acts as a digital mailing address for a device.
- **Domain Name System (DNS):** Translates human-readable web addresses into numerical IP addresses.
- **Latency:** Time it takes for a data packet to travel from its source to its destination.
- **Bandwidth:** The maximum amount of data that can be transmitted over a network connection in a given amount of time.

Google Cloud's network



Regions: Independent geographic areas containing multiple zones.



Zones: Isolated locations within regions to protect against single-failure events.



Edge locations: Globally distributed points of presence that cache content near users.

Open standard versus open source

- **Open standard:** A publicly available specification for a technology (for example, TCP/IP and SQL). By providing a common language or framework, open standards prevent vendor lock-in and ensure that diverse systems can communicate seamlessly.
- **Open source:** Software for which the source code is made publicly available and can be inspected, modified, and enhanced by anyone (for example, Linux and PyTorch). This model encourages rapid innovation, as a global community of developers can contribute to the software.